

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

heart = pd.read_csv(r'C:\Users\751pa\Downloads\heart_disease.csv')

print("Available columns in the dataset:")
print(heart.columns)

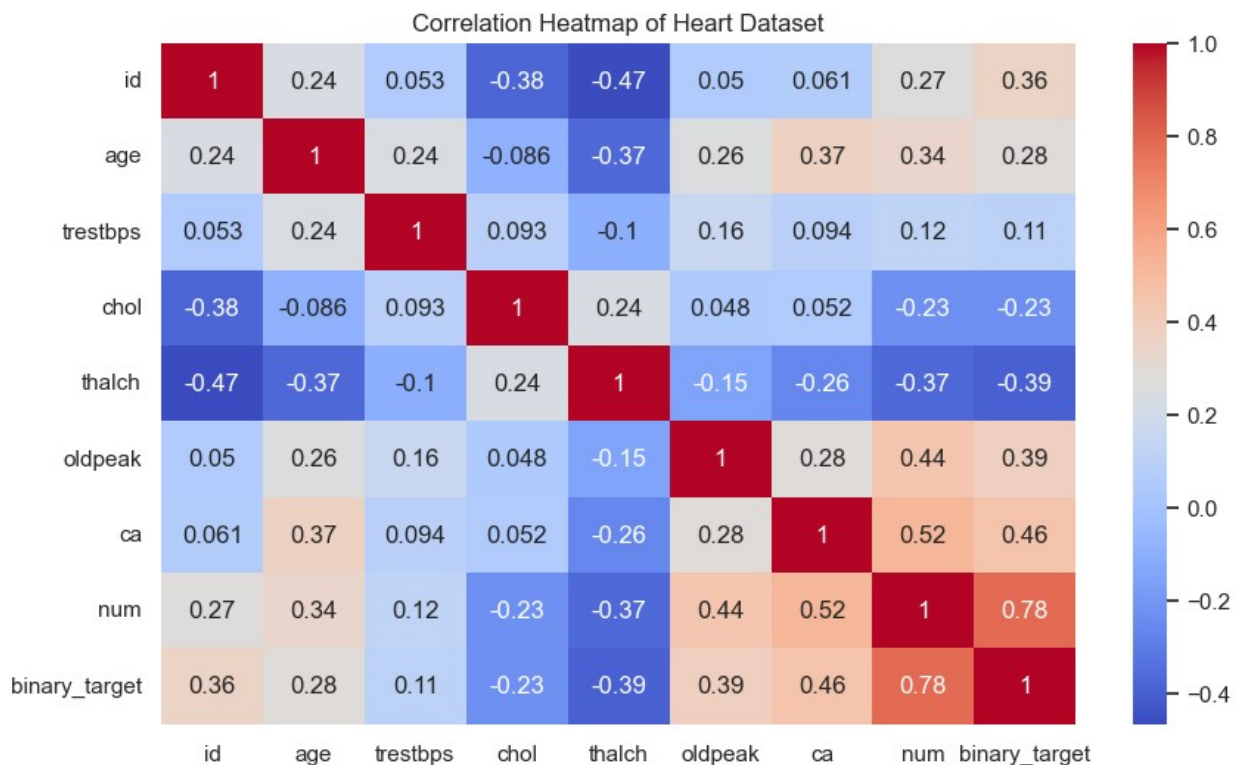
Available columns in the dataset:
Index(['id', 'age', 'sex', 'dataset', 'cp', 'trestbps', 'chol', 'fbs',
       'restecg', 'thalch', 'exang', 'oldpeak', 'slope', 'ca', 'thal',
       'num'],
      dtype='object')

heart['binary_target'] = heart['num'].apply(lambda x: 0 if x == 0 else 1)

sns.set(style="whitegrid")

plt.figure(figsize=(10, 6))
sns.heatmap(heart.corr(numeric_only=True), annot=True,
            cmap='coolwarm')
plt.title("Correlation Heatmap of Heart Dataset")
plt.show()

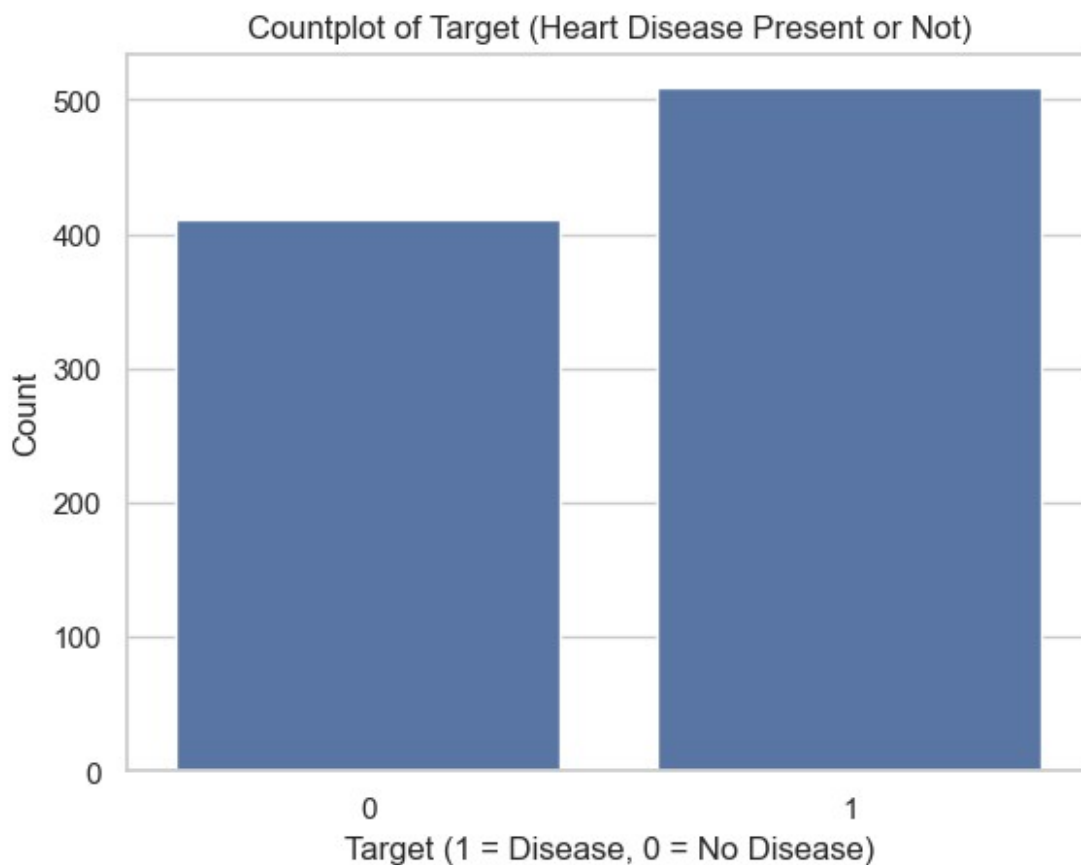
```



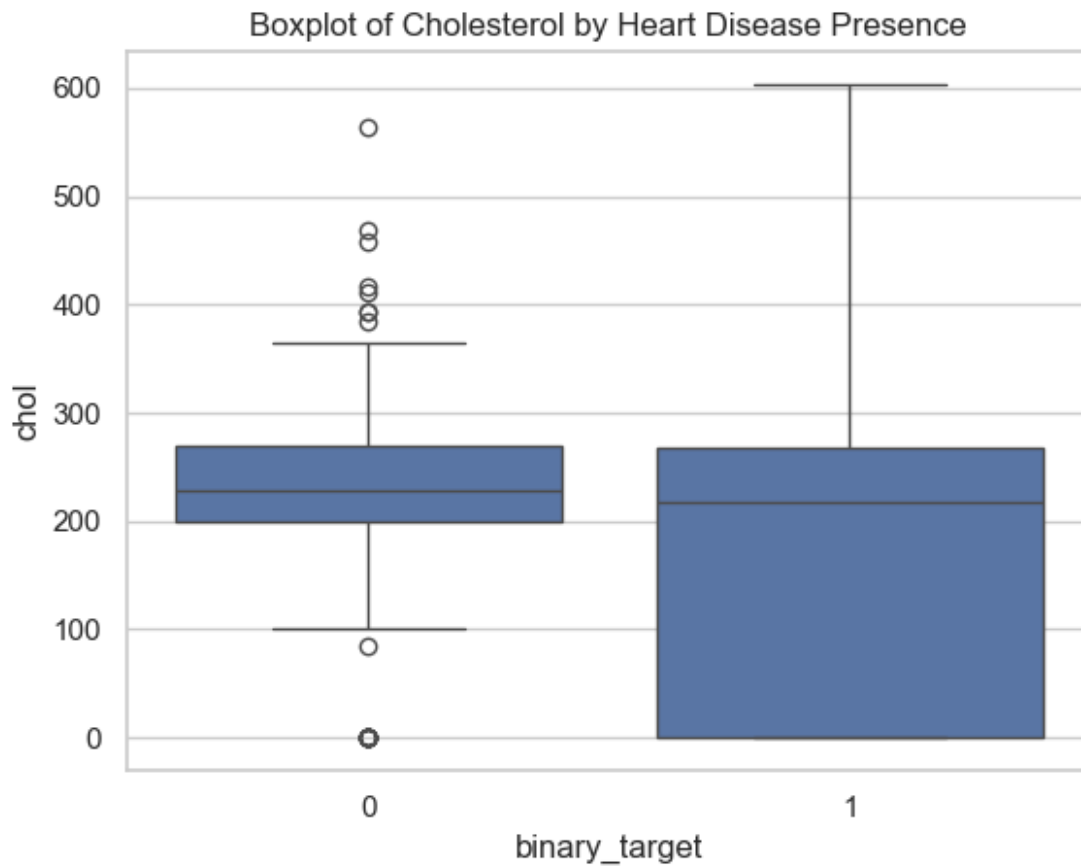
```
heart.columns

Index(['id', 'age', 'sex', 'dataset', 'cp', 'trestbps', 'chol', 'fbs',
      'restecg', 'thalch', 'exang', 'oldpeak', 'slope', 'ca', 'thal',
      'num',
      'binary_target'],
      dtype='object')

sns.countplot(x='binary_target', data=heart)
plt.title("Countplot of Target (Heart Disease Present or Not)")
plt.xlabel("Target (1 = Disease, 0 = No Disease)")
plt.ylabel("Count")
plt.show()
```

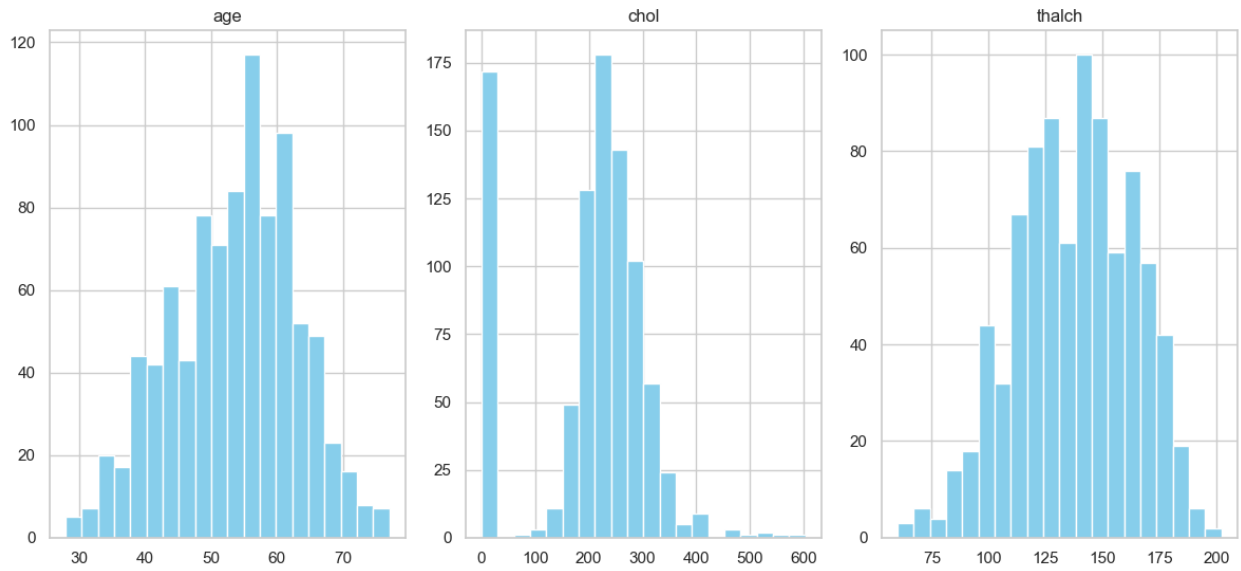


```
sns.boxplot(x='binary_target', y='chol', data=heart)
plt.title("Boxplot of Cholesterol by Heart Disease Presence")
plt.show()
```

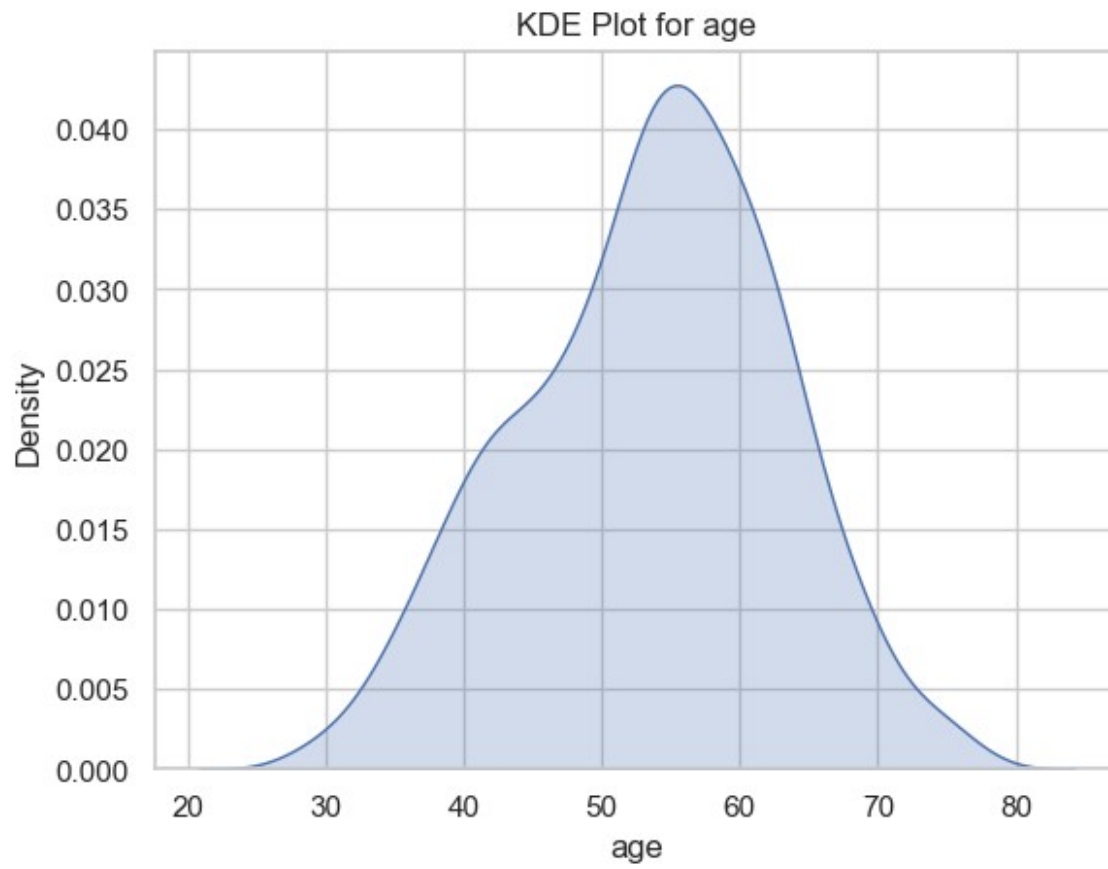


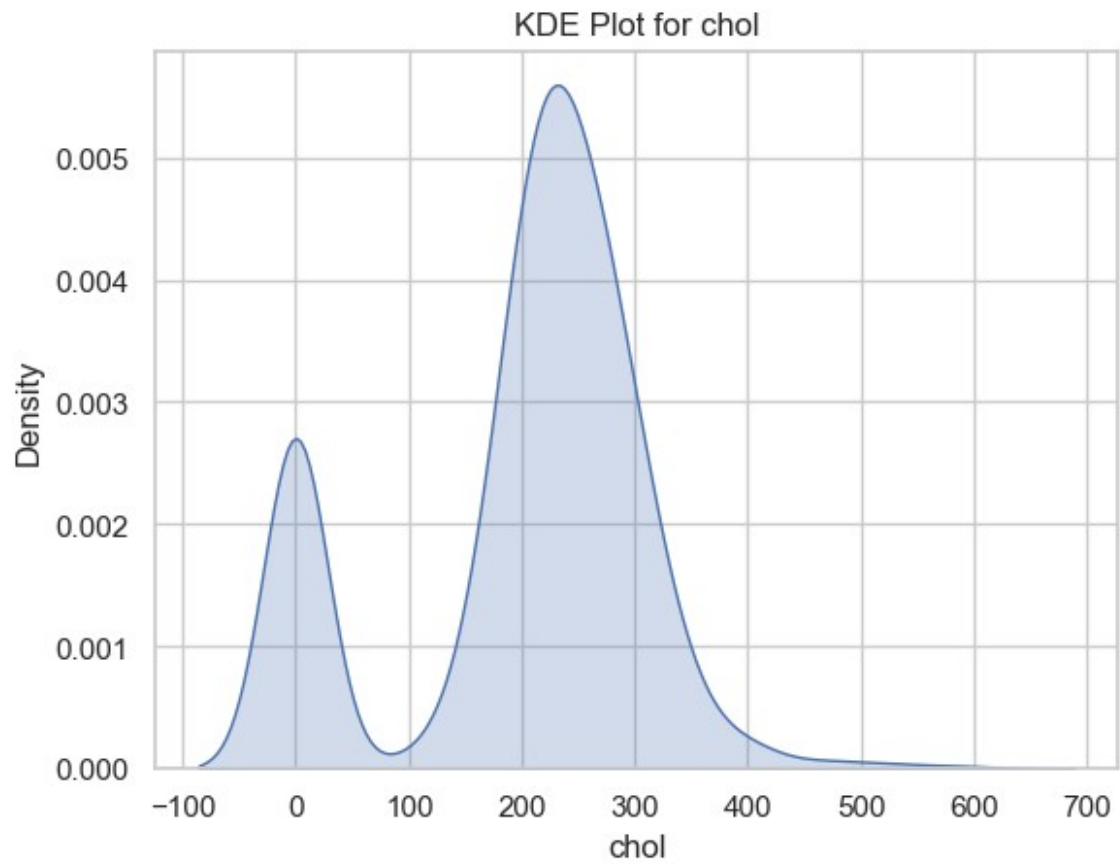
```
heart[['age', 'chol', 'thalch']].hist(bins=20, figsize=(12, 6),
layout=(1, 3), color='skyblue')
plt.suptitle("Histograms of Age, Cholesterol, and Max Heart Rate")
plt.tight_layout()
plt.show()
```

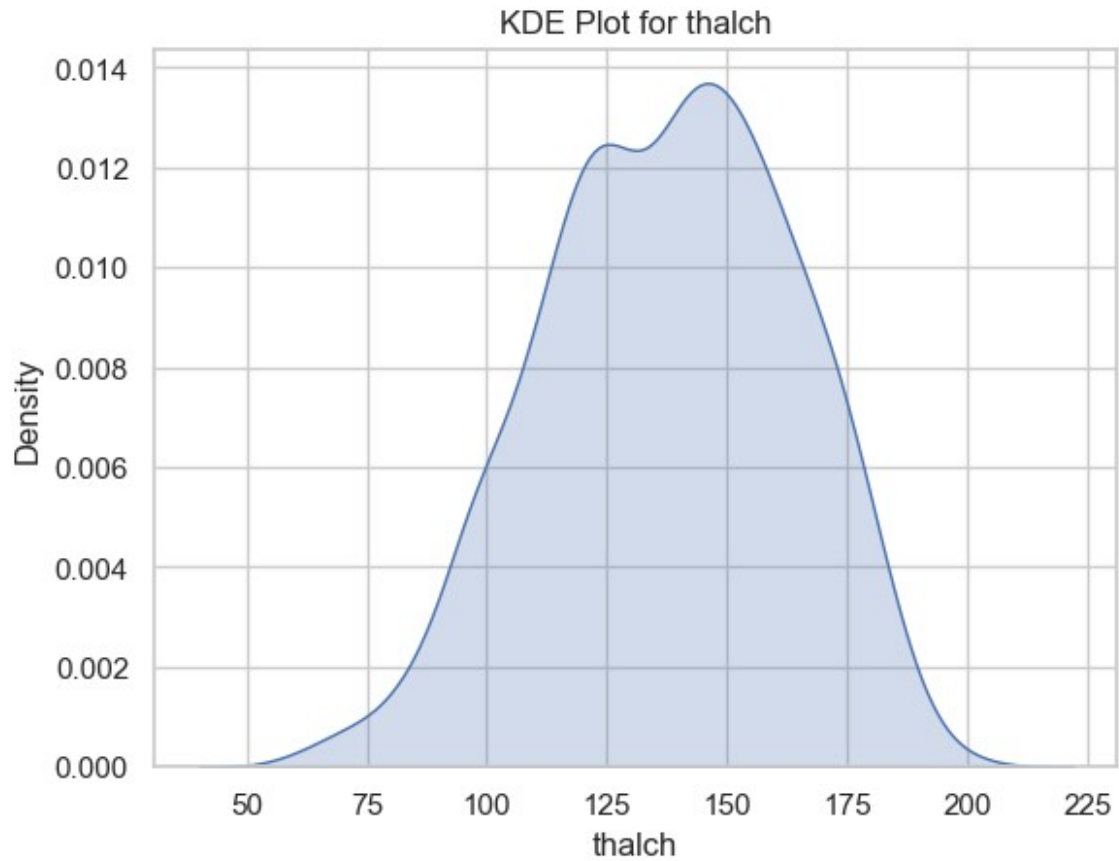
Histograms of Age, Cholesterol, and Max Heart Rate



```
for column in ['age', 'chol', 'thalch']:  
    sns.kdeplot(data=heart[column], fill=True)  
    plt.title(f"KDE Plot for {column}")  
    plt.xlabel(column)  
    plt.show()
```







```
sns.jointplot(data=heart, x='age', y='chol', kind='scatter',  
hue='binary_target')  
plt.suptitle("Jointplot of Age vs Cholesterol", y=1.02)  
plt.show()
```

Jointplot of Age vs Cholesterol

